Emergent Riemannian Topology from a Pre-Geometric Graph Substrate: A Variational Matrix Toy Model

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Abstract

We introduce a minimal, background-independent numerical framework where continuous Riemannian geometry emerges from a discrete, relational network substrate. The pre-geometric state is modeled via a symmetric adjacency matrix governed by an action-minimizing variational Matrix Hamiltonian. By establishing a fully permutation-invariant Hamiltonian utilizing spectral trace properties, we resolve fundamental gauge ambiguities common to discrete topological models. Using Markov Chain Monte Carlo (MCMC) simulations, we confirm that the framework avoids non-physical coordinate dependence while dynamically driving random graphs toward homogeneous, isotropic spatial manifolds. Finally, we implement a conservation-law-preserving node insertion mechanism to simulate steady spatial expansion, offering a microphysical toy model for uniform cosmic inflation.

1 Introduction

A primary obstacle in modern quantum gravity frameworks is background dependence, where quantum field operators are conventionally defined on a fixed, smooth spacetime container. An alternative paradigm treats smooth continuous geometry as an emergent macroscopic aggregate of underlying discrete, purely relational degrees of freedom.

In this work, we present a minimal background-independent framework where spatial manifolds co-emerge from a graph substrate. Unlike causal set theory, which enforces strict partial ordering, our model utilizes continuous weighted links. Unlike prior quantum graphity formulations, we enforce strict gauge invariance via a permutation-invariant variational Hamiltonian. We detail the exact continuum limit required to recover a standard Riemannian metric signature and test the model's structural growth under localized graph strain.

2 Mathematical Formalism

Let the pre-geometric substrate Q be represented by a weighted, symmetric adjacency matrix $A_{ij} \in [0, 1]$ mapping N fundamental relational nodes. The discrete spatial architecture is investigated via the symmetric Graph Laplacian operator:

$$\Delta f_i = \sum_j A_{ij}(f_i - f_j) \tag{1}$$

The network relaxes toward geometric equilibrium via the minimization of a variational Matrix Hamiltonian:

$$\mathcal{H}[Q] = \alpha\mathcal{H}_{\text{dim}} + \beta\mathcal{H}_{\text{iso}} + \gamma\mathcal{H}_{\text{reg}} \tag{2}$$

where α, β, γ represent positive scaling constants balancing competing topological energy penalties.

2.1 Permutation-Invariant Energy Penalties

To satisfy fundamental gauge covariance, the Hamiltonian must be invariant under arbitrary rearrangements of the node labels (permutation matrices P).

2.1.1 Dimensional Penalty

The spectral dimension d_s is dynamically monitored via the heat kernel trace over a diffusion time t:

$$d_s(t) = \frac{2t\sum_k \lambda_k e^{-\lambda_k t}}{\sum_k e^{-\lambda_k t}} \tag{3}$$

where λ_k are the eigenvalues of the Graph Laplacian. The dimensional energy penalty is formalized as:

$$\mathcal{H}_{\text{dim}} = (d_s(t) - D_{\text{target}})^2 \tag{4}$$

where we fix the target macroscopic dimension to $D_{\text{target}} = 3$.

2.1.2 Isotropy Measure

To enforce spatial homogeneity and isotropic connectivity across all nodes, we minimize the structural variance of the node degrees ($k_i = \sum_j A_{ij}$) alongside the spectral variance:

$$\mathcal{H}_{\text{iso}} = \frac{1}{N}\sum_i (k_i - \bar{k})^2 + \frac{1}{N}\sum_k (\lambda_k - \bar{\lambda})^2 \quad (5)$$

2.1.3 Structural Regularizer

To penalize sharp geometric discontinuities without breaking node-labeling gauge freedom, we implement the trace of the squared Laplacian as a structural regularizer:

$$\mathcal{H}_{\text{reg}} = \frac{1}{N}\text{Tr}(\Delta^2) = \frac{1}{N}\sum_i \lambda_i^2 \quad (6)$$

Because the matrix trace is cyclic ($\text{Tr}(P^T\Delta^2 P) = \text{Tr}(\Delta^2)$), $\mathcal{H}_{\text{reg}}$ is identically independent of node index ordering, resolving the gauge issues of prior index-dependent formulations.

2.2 Continuum Limit and Spatial Emergence

We map a continuous spatial scalar field $\phi(x)$ to the discrete node coordinates via $f_i = \phi(x_i)$. Taylor expanding a localized neighborhood to second order yields:

$$\phi(xj) = \phi(xi) + \Delta x^\mu \partial_\mu \phi + \frac{1}{2}\Delta x^\mu \Delta x^\nu \partial_\mu \partial_\nu \phi + \mathcal{O}(a^3) \quad (7)$$

where a is the characteristic lattice spacing. Substituting this expansion back into the discrete Graph Laplacian yields:

$$\Delta\phi = -\left(\sum_j A_{ij}\Delta x^\mu\right)\partial_\mu\phi - \frac{1}{2}\left(\sum_j A_{ij}\Delta x^\mu \Delta x^\nu\right)\partial_\mu\partial_\nu\phi \quad (8)$$

The structural minimization of $\mathcal{H}_{\text{iso}}$ drives the network configuration to satisfy:

$$\sum_j A_{ij}\Delta x^\mu = 0 \quad (9)$$

$$\sum_j A_{ij}\Delta x^\mu \Delta x^\nu \to a^2 \delta^{\mu\nu} \quad (10)$$

where $\delta^{\mu\nu}$ is the positive-definite Kronecker delta. Thus, as $a \to 0$, the graph Laplacian cleanly converges to the continuous Riemannian Laplace-Beltrami operator:

$$\lim_{a\to 0} \frac{2}{a^2}\Delta = \nabla^2 \quad (11)$$

3 Numerical Simulation & Matrix Growth

The model is executed using a Metropolis-Hastings Markov Chain Monte Carlo (MCMC) simulated annealing routine. In the static network regime, the system converges reliably toward a homogeneous spatial lattice, stabilizing around the target spectral dimension.

3.1 Strain-Driven Node Insertion

To simulate an expanding background, we present a phenomenological node-generation mechanism driven by local bond density. When a highly localized relational connection exceeds a critical density threshold ($A_{ij} > \kappa_{\text{crit}} = 0.80$), a new node N_{new} is dynamically inserted to split the link.

To prevent unphysical energy shocks during a birth event, the adjacency entries are scaled by a preserving fraction $\zeta = 0.7$:

$$A'_{i,N_{\text{new}}} = A'_{N_{\text{new}},i} = A_{ij}\zeta \quad (12)$$

$$A'_{j,N_{\text{new}}} = A'_{N_{\text{new}},j} = A_{ij}\zeta \quad (13)$$

In our toy model simulation initialized with $N_{\text{start}} = 80$, introducing highly dense core seed regions successfully triggers continuous node generation, scaling the network uniformly to $N = 116$ before stabilizing into a expanded, relaxed topology.

4 Discussion & Limitations

By restricting the framework to a symmetric adjacency matrix ($A_{ij} = A_{ji}$) and a positive semi-definite Graph Laplacian, the model successfully avoids the metric signature conflicts found in early pre-geometric formulations. However, because a Riemannian framework lacks an explicit time dimension, this configuration represents a static spatial slice or an Euclidean quantum gravity formulation rather than a dynamic spacetime.

Future extensions will explore moving from a symmetric matrix to a complex-valued, pseudo-Hermitian adjacency architecture ($A_{ij} \in \mathbb{C}$), which could allow a Lorentzian metric signature ($-{+}{+}{+}$) to emerge via the secondary phase structures of the network trace. Additionally, scaling the system beyond the current toy boundaries ($N \leq 250$) is required to evaluate whether the scaling relation $a(t) \propto N^{1/3}$ remains valid for macro-systems.

5 Conclusion

This revised framework provides a mathematically consistent, permutation-invariant toy model for emergent Riemannian geometry. By replacing coordinate-dependent strain equations with invariant spectral traces, we demonstrate that a discrete matrix network can seamlessly relax into a smooth, isotropic spatial manifold. This establishes a robust numerical substrate for exploring background-independent physics.